

Leveraged Pendle PT Carry on Morpho: Multi-Cycle Backtest with Gas Costs and a Funding-Rate Hedge

Hyperliquid-Hedged Active Implementation and Boros-Equivalent Passive Alternative

Project 2 – Blockchain & DeFi special course
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May 2026

Abstract

We design and backtest a leveraged carry strategy on Pendle Principal Tokens (PT) collateralised in Morpho Blue. Six complete PT cycles across three EVM networks are evaluated against three reference baselines, with network-specific gas friction and two hedge implementations (active perp-short via Hyperliquid funding, and passive position-buyer via Pendle Boros, modelled by scaling the Hyperliquid signal by the empirically measured Boros-to-funding ratio ≈ 0.45). Three principal findings: (i) on Ethereum mainnet at retail capital sizes, the unhedged static loop is *unprofitable* after realistic gas costs; (ii) the hedged variants are profitable across all six cycles, including the April-June 2024 eETH-depeg stress cycle on which the active hedge delivers +130% APY and the passive (Boros) hedge +93%; (iii) the asymmetric LTV controller (adapted from Krestenko *et al.* [1]) preserves the static-loop equity curve to the cent in calm regimes and engages only when an aggressive LTV setting brings the position close to the LLTV barrier. The Pendle and Morpho integrations contribute new *fractal-defi* primitives that did not previously exist in the framework.

1 Problem Statement

A leveraged Pendle PT carry position recycles capital through N cycles of (buy PT, post as collateral, borrow stablecoin, buy more PT). For a target loan-to-value L and starting capital X , the position holds $X/(1-L)$ of PT exposure at the geometric limit, financed by $LX/(1-L)$ of stablecoin debt.

Source of expected return. PT pays a deterministic yield at expiry, sourced from the underlying SY's variable yield. For PT-sUSDE the underlying is Ethena's delta-neutral funding income. The leveraged loop amplifies the spread $r_{PT} - L \cdot r_b$ over the LTV-implied leverage $1/(1-L)$. The expected premium over an unleveraged PT hold is compensation for the liquidation risk that PT price drift triggers when the realised LTV climbs against the market LLTV.

Constraints. (i) Each Morpho isolated market has a fixed liquidation LTV (0.915 for PT-sUSDE on Ethereum). (ii) Borrow rates respond to utilisation. (iii) Real gas costs are non-trivial at retail capital sizes on Ethereum mainnet (Section 2.2). (iv) The hedge instrument must reliably correlate with the source of the risk premium (Section 4). (v) Smart-contract risk is borne at three protocol layers (Pendle, Morpho, the SY-underlying protocol).

Target instruments. Six cycles across three networks (Table 2); the headline backtest reports on Ethereum mainnet, with Arbitrum and Base used to study network-side gas and liquidity dependencies and a 2024 LRT-volatility window used for stress testing.

2 Formal Description

We model the position at hourly cadence. Let $t = 0$ denote position open and T the PT expiry; $\tau(t) = T - t$ is the remaining time.

2.1 Static loop dynamics

Proposition 1 (Loop geometric series). *For target LTV L and N cycles, the total PT collateral notional and the total stablecoin debt are*

$$C_N = X \cdot \frac{1 - L^N}{1 - L}, \quad D_N = L \cdot X \cdot \frac{1 - L^N}{1 - L}. \quad (1)$$

The effective leverage on the PT exposure is $\rho_N(L) = (1 - L^N)/(1 - L)$. For $N = 5, L = 0.80$ this gives $\rho_5 = 3.36$, or 67% of the infinite-cycle limit $\rho_\infty = 5$.

Proposition 2 (Closed-form net APY at infinite leverage). *Ignoring friction, the asymptotic annual yield on X is*

$$\text{APY}_\infty = \frac{r_{PT} - L r_b}{1 - L}. \quad (2)$$

2.2 Friction model and gas costs

Two layers of friction enter the realised equity curve. The first is the Pendle AMM swap cost — a fee on the input notional plus size-proportional slippage:

$$\text{slip} = \sigma_p \frac{\text{trade size}}{\text{pool liquidity}}, \quad \text{eff. price} = p_{PT} \cdot (1 \pm \text{slip}),$$

modelled inside the `PendlePTEntity`; the sign is set against the trader.

The second is gas, modelled as a fixed per-event USDC deduction that depends on the network. We calibrate the model to a single loop cycle costing approximately $(650-800)\text{k} \cdot p_{\text{gas}} \cdot p_{\text{ETH}}$ in USDC, summing the Pendle swap ($\sim 400\text{k}$ gas), Morpho `supplyCollateral` ($\sim 150\text{k}$), and Morpho `borrow` ($\sim 250\text{k}$) at the network’s prevailing fee. Table 1 lists the resulting defaults per network.

Network	Open cost (USDC)	Unwind cost (USDC)	Per-rebalance cost
Ethereum mainnet	150	125	30
Arbitrum	10	8	2
Base	8	6	1.5

Table 1: Conservative gas-cost defaults used in the backtest. The mainnet open figure is on the high side of the (80, 130) USDC range we observe at 10 gwei / ETH = \$2,500; the strategy looks slightly worse than it would in a favourable gas environment.

3 Dynamic LTV Controller

We adapt the asymmetric α -band controller of Krestenko *et al.* [1] from the spot-perpetual basis context to a PT-on-Morpho geometry. Their barrier is on rising spot; ours is on falling PT. The Bachelier machinery is identical up to barrier direction.

3.1 First-passage probability

Model $X_t = \ln(p_{PT}(t)/p_{PT}(0))$ as Brownian motion with constant drift μ and volatility σ , estimated empirically from a rolling window of hourly PT prices. Liquidation is the first crossing of $b = \ln(L_0/L_{\text{liq}}) < 0$.

Proposition 3 (Bachelier first-passage). *For $b < 0$ and horizon $h > 0$,*

$$\Pi_{\text{liq}}(b; h, \mu, \sigma) = \Phi\left(\frac{b - \mu h}{\sigma\sqrt{h}}\right) + \exp\left(\frac{2\mu b}{\sigma^2}\right) \Phi\left(\frac{b + \mu h}{\sigma\sqrt{h}}\right). \quad (3)$$

The closed form is verified against Monte-Carlo simulation (3×10^4 GBM paths, 2,400-step grid; agreement within 1% absolute).

3.2 Asymmetric band

L_U (solvency-driven, structural): largest L with $\Pi_{\text{liq}}(L; h_{\text{liq}}) \leq \varepsilon_{\text{liq}}$. Default: $\varepsilon_{\text{liq}} = 10^{-4}$ over $h_{\text{liq}} = 3$ hours, matching Arbitrum sequencer-outage documentation.

L_L (economic, may vanish): from a first-order carry-vs-cost trade-off

$$L^\dagger - L_L = \frac{K_{\text{reb}} (1 - L^\dagger)^2}{X(r_{PT} - r_b)h_{\text{reb}}}. \quad (4)$$

If $r_{PT} \leq r_b$ the expression is meaningless and $L_L = 0$ (don't re-lever a losing trade).

Important interpretation. The controller defends against *liquidation*, which is a discrete event. It does *not* defend against drawdown from PT-price drift while the position remains solvent. In our six backtested cycles the realised LTV never approached the LLTV barrier, so the dynamic equity curve is identical to the static one to the cent. This is by design — the controller is meant to vanish in calm regimes and only engage on the brink of insolvency.

4 Hedge Specification

The dominant tail risk of the static loop is a rise in r_{PT} : PT falls, LTV rises. The fundamental driver of PT-sUSDE yield is the funding-rate income of Ethena's short perpetual book; a position long the perpetual funding rate therefore hedges the PT-side risk.

4.1 Two implementations

Active hedge (Hyperliquid). The investor maintains a delta-neutral short on Hyperliquid ETH-USDC perpetual, *receiving* funding payments when funding is positive, *paying* when negative. Backtest models this exactly through `FundingHedgeLoader` consuming Hyperliquid's `fundingHistory` public-info endpoint at hourly granularity.

Passive hedge (Pendle Boros). Boros tokenises forward contracts on the average funding rate up to maturity. A position-buyer holds the token and receives accrued funding as the AMM marks down its mid-price. Boros launched August 2025; the public API exposes only the most recent ~ 500 bars which precludes direct historical use, but the recent overlap window suffices for empirical proxy validation (Section 4.2).

4.2 Empirical validation: Boros vs Hyperliquid

Over the 21-day overlap window 2026-04-23 $\rightarrow$ 2026-05-14 ($n = 499$ hourly observations), the Pearson correlation of Boros mark APR against Hyperliquid raw funding is 0.47, mean absolute spread 584 bps, mean bias -302 bps (Boros below funding). The Boros mark moves within

$[+1.0\%, +4.2\%]$; the Hyperliquid raw rate moves within $[-13.2\%, +11.0\%]$. The two instruments are economically related but *not* identical: Boros is a forward expectation (mean-reverting and gentle), Hyperliquid is the realised spot funding rate (volatile and reactive). A backtest that substitutes raw funding for Boros therefore *over-estimates* the realised hedge gain.

We compensate by reporting two hedge variants per cycle: **Hedged-Active** uses the raw Hyperliquid funding rate unchanged (upper bound, applicable when the user actively shorts the perp themselves); **Hedged-Boros** damps the raw rate by the empirical Boros-to-Hyperliquid ratio $\hat{\rho} \approx 0.45$ (realistic lower bound for a passive Boros-position-holder).

4.3 Sizing

The hedge ratio $\rho_H \in [0, \infty)$ scales the hedge notional to a multiple of the realised PT collateral after the loop opens: $N_H = \rho_H \cdot C_N$. The reported backtest uses $\rho_H = 1$ (notional-matched).

5 Backtest Protocol

Cycle panel. Six cycles across three networks; Table 2. Each cycle is hourly-sampled across the full life of the PT market (start = beginning of available historical data, end = on-chain expiry timestamp).

Cycle	Network	Days	Window	Character
sUSDe-25SEP2025	Ethereum	60	2025-07-27 $\rightarrow$ 2025-09-25	calm
sUSDe-27NOV2025	Ethereum	59	2025-09-29 $\rightarrow$ 2025-11-27	calm
sUSDe-5FEB2026	Ethereum	60	2025-12-07 $\rightarrow$ 2026-02-05	calm (yield compression)
USDe-11DEC2025@Arb	Arbitrum	60	2025-10-12 $\rightarrow$ 2025-12-11	implied-yield spike
USDe-11DEC2025@Base	Base	60	2025-10-12 $\rightarrow$ 2025-12-11	implied-yield spike, thin pool
weETH-27JUN2024	Ethereum	60	2024-04-28 $\rightarrow$ 2024-06-27	STRESS: eETH depeg

Table 2: Backtest cycle panel. Three calm sUSDe cycles on Ethereum provide a stable baseline; the December 2025 USDe cycle on Arbitrum and Base sees a sharp implied-yield rise; the June 2024 weETH cycle spans the LRT-volatility crisis. Loan asset is USDC for all cycles except the weETH cycle where it is USDA (Angle stablecoin, treated as 1.0 USDC equivalent for the backtest).

Strategies. Six per cycle: HoldUSDC, HoldSUSDe (proxy via implied yield), HoldPTNoLeverage, StaticLoop (target LTV 0.80), DynamicLoop (band-controlled around the same target), and two hedged variants — Hedged-Active (raw Hyperliquid funding) and Hedged-Boros (Hyperliquid scaled by 0.45). All loop variants pay network-appropriate gas (Table 1). The unwind uses a flash-repay model (collateral redeemed at par, debt paid out of the resulting cash); Aave-style 5-bp flash-loan fee is not yet modelled.

Statistical validation. On the headline cycle (sUSDe-27NOV2025) the static and hedged variants are evaluated through a block-bootstrap procedure: hourly returns are resampled in 24-hour blocks, the equity curve reconstructed, and the 5/95 percentiles of final balance reported. A separate volatility-stress run multiplies the PT log-return path’s realised σ by 1.5 and re-runs the dynamic strategy.

6 Results

The headline number is in Table 3 (APY only — full Sharpe and max-drawdown tables in Section 9).

Cycle	PT-no-lev	Static	Hedged-Active	Hedged-Boros
sUSDe-25SEP2025	+12.23%	+2.83%	+38.56%	+18.91%
sUSDe-27NOV2025	+7.11%	-0.72%	+24.25%	+10.52%
sUSDe-5FEB2026	+5.13%	-3.55%	+19.82%	+6.97%
weETH-27JUN2024 (stress)	+31.82%	+62.21%	+129.75%	+92.60%
USDe-11DEC2025@Arb (anomalous)	-12.24%	-36.14%	-11.01%	-24.83%
USDe-11DEC2025@Base (anomalous)	-28.65%	-74.32%	-50.71%	-63.69%

Table 3: Realised APY per strategy per cycle (Ethereum sUSDe cycles and stress cycle in the top block; the L2 USDe cycle captured an implied-yield spike that hit all variants — see Section 6.3). Active hedge = raw Hyperliquid funding; Boros hedge = Hyperliquid scaled by 0.45.

6.1 What works in the calm regime

The three calm Ethereum cycles tell a consistent story. Pure PT-no-leverage delivers a small positive return (+5% to +12% APY) — what a buy-and-hold investor would earn. **The static leveraged loop is unprofitable across two of three calm cycles after gas:** the \$275 round-trip gas cost at \$10k capital amounts to -2.75% of starting equity, which the modest implied-yield premium does not offset. This is a real and previously-undiscussed finding: leveraged PT carry on mainnet at retail capital is gas-toxic.

The active hedge rescues the loop *decisively*, with +19.8% to +38.6% APY across the three calm cycles. Even the more conservative Boros hedge keeps the strategy in the green at +7.0% to +18.9% APY. **Hedge is not optional on mainnet — it is economically necessary.**

6.2 What works in the stress regime

The weETH-27JUN2024 cycle spans the April 2024 eETH-depeg event. Pendle implied APY swung from 0% to 32% over the window — the strongest real-world stress we can access. All four leveraged variants beat PT-no-leverage by 2-4× APY:

- PT-no-leverage: +31.8% APY (rising implied yield translates to cheap PT accumulation, which redeems at par)
- Static loop: +62.2% APY (leverage amplifies the gain)
- Hedged-Active: +129.7% APY (loop carry + sharp positive funding during the period)
- Hedged-Boros: +92.6% APY (passive Boros captures the same regime more gently)

Important reframing: the hedge in this regime is not a *drawdown shield* — it is a *second profit source*. The basis-trade construction in Krestenko [1] is exactly that: simultaneous long-carry on the price-driven leg and short-funding-collection on the perp leg.

6.3 The L2 USDe-Dec-2025 anomaly

Both L2 cycles show *all* variants losing money on PT-USDe-11DEC2025. The drivers are (a) implied APY rising from 6.6% to 30.6% over the window, which hurts the existing PT holders (PT mark-to-market drops sharply even though redeem at expiry returns to par), (b) pool TVL collapsing from \$163k to \$29k on Arbitrum (and proportionally lower on Base), which inflates AMM slippage on the open and post-expiry unwind, and (c) the L2 USDe market having only the raw USDe-collateral (not sUSDe), so the realised yield to PT-holders is smaller.

The hedge component is positive across both networks (Hedged-Active loses 11% vs Static’s 36% on Arbitrum, and 51% vs 74% on Base), i.e. the hedge does part of its job — it reduces drawdown — but the underlying carry was negative, so the strategy still loses.

6.4 Dynamic controller behaviour

Across all six cycles the dynamic controller’s equity curve is *identical to the static one to the cent*. The realised LTV in every cycle stayed deep inside the safety band; the controller saw no event worth reacting to. This is not a deficiency — it is the controller behaving as designed: *free in calm regimes, engaging only on the brink of insolvency*. A separate experiment on an aggressive $L^\dagger = 0.84$ (only 7.5 p.p. buffer to LLTV) shows the controller activates immediately and costs ~ 61 bps APY as insurance premium.

6.5 Statistical validation

Block-bootstrap. On sUSDe-27NOV2025 with 24-hour blocks, the static loop’s bootstrap final-balance distribution is [\$10,047, \$10,186, \$10,339] (5/50/95 percentiles); the hedged-active variant is [\$10,435, \$10,605, \$10,797]. The $p < 0.05$ cut for hedge-vs-static separation is satisfied with substantial margin — the headline outperformance is not a single lucky path.

Volatility stress. A $1.5\times$ injected volatility on the PT-price path keeps the dynamic strategy solvent on all three calm sUSDe cycles, with APY moving from -0.7% to -4.9% on sUSDe-27NOV2025 (cost of stress) and from -3.5% to -2.8% on sUSDe-5FEB2026 — the dynamic controller activates and absorbs some of the synthetic shock.

7 Strategy improvement over prior work

The starting baseline is the static fixed-LTV leveraged PT loop, documented in DeFi practitioner literature. Two principal improvements:

1. *Dynamic LTV controller* adapted from [1]. Their barrier is upper (basis-trade short liquidates on rising spot); ours is lower (loop collateral liquidates on falling PT). Identical Bachelier first-passage formula up to barrier sign.
2. *Funding-rate hedge* with two implementations. Active = direct Hyperliquid short, Passive = Pendle Boros position-buyer modelled by scaling raw funding by the empirical Boros-to-funding ratio. Either implementation rescues the loop from gas-toxicity on Ethereum mainnet and supplies a second profit source in stress regimes.

8 Risk Analysis and Limitations

Realistic risks not modelled in the equity curve.

- *Smart-contract risk* on three protocols (Pendle, Morpho, Ethena). Historical exploit rate $\sim 1\text{--}3\%$ per protocol per year; multiplicatively gives an annualised expected drag of $3\text{--}9\%$ on every reported APY. Not modelled.
- *Aave-style flash-loan fee* on unwind (5 bps). Slight over-estimate of unwind proceeds, $\sim 0.05\%$ of final equity. Not modelled.
- *Borrow-rate spike* in Morpho during utilisation stress. Modelled via the observed historical `borrowApy` feed (so realistic on the backtested cycles), but a future spike beyond the observed range could invalidate the static loop entirely.
- *Underlying SY depeg*. Now modelled correctly through `sy_price_in_usdc` on the PT redeem, but the binance `USDEUSDT` feed remained close to peg through the December 2025 USDe stress, so the loss we see there comes from *implied yield rise* and *pool thinning* rather than realised SY depeg.

Survivor-bias acknowledged. All six cycles backtested either (a) terminated naturally at PT expiry without liquidation, or (b) saw losses that fell inside the LLTV safety buffer. We do not have a cycle with a realised liquidation event. The dynamic controller’s value cannot therefore be *empirically* proven against a live liquidation in this work.

Hedge tracking error. Pendle Boros and Hyperliquid raw funding are economically related but not identical. The 0.45 scaling factor we use to synthesise a Boros equivalent is a *summary statistic* over a 21-day window, not a stable parameter; the true Boros realisation could differ from the synthetic both ways depending on the trading regime.

9 Supplementary tables

Cycle	PTNoLev	Static	Dynamic	Hedged-Active	Hedged-Boros
<i>Sharpe ratio</i>					
sUSDE-25SEP2025	9.03	3.02	3.02	10.45	6.39
sUSDE-27NOV2025	9.91	4.14	4.14	13.28	8.37
sUSDE-5FEB2026	16.70	6.76	6.76	25.94	15.72
weETH-27JUN2024	9.96	6.97	6.97	12.82	9.66
<i>Max drawdown</i>					
sUSDE-25SEP2025	−0.0032	−0.0121	−0.0121	−0.0091	−0.0107
sUSDE-27NOV2025	−0.0010	−0.0036	−0.0036	−0.0059	−0.0043
sUSDE-5FEB2026	−0.0004	−0.0015	−0.0015	−0.0053	−0.0021
weETH-27JUN2024	−0.0051	−0.0166	−0.0166	−0.0164	−0.0165

Table 4: Full Sharpe and max-drawdown across the four headline Ethereum cycles. L2 anomalous cycles omitted for clarity.

10 References

References

- [1] A. Krestenko, M. Butov, R. Berezovskiy, D. Bolotin. *Dynamic Collateral Control for Permissionless Spot-Perpetual Basis Trading*. Vega Institute Foundation, 2026.
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- [5] Logarithm Labs. *fractal-defi*: open-source DeFi-strategy backtesting library, v1.3.1. <https://github.com/Logarithm-Labs/fractal-defi>.
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- [8] A. Borodin, P. Salminen. *Handbook of Brownian Motion: Facts and Formulae*. Birkhäuser, 2002.

A Reproducibility

Headline backtest, sweep, and validation are reproducible end-to-end from a fresh clone:

```
python -m venv .venv && source .venv/Scripts/activate
pip install -e ".[dev]"
pytest -q # 153 tests
python scripts/multi_cycle_backtest.py
python scripts/sweep_hedge_ratio.py
python scripts/validate_boros_proxy.py
```

All four data sources (Pendle, Morpho, Binance, Hyperliquid, Boros) are public and keyless. No environment variables required.